Disparities in COVID-19 Infection in San Francisco: A Bayesian Latent Time Trend Model

Keith Hall
jkhall@gatech.edu

***Abstract*—**This report investigates disparities in COVID-19 infection risk across racial and ethnic groups in San Francisco using hierarchical Bayesian Negative Binomial (NB) models. Publicly available case data were modeled using a log-link with group-level effects and a Gaussian random walk that captures broad changes in infection intensity. Posterior contrasts between group effects indicate that Native Hawaiian or Other Pacific Islander and Hispanic/Latino residents experienced substantially higher per-capita infection risk than white and Asian residents, with Black/African American residents also elevated relative to whites. A second model incorporating era-specific effects and group–era interaction terms was used to explore the evolution of these disparities across three eras of the pandemic (pre-vaccine, vaccine/Delta, and Omicron-and-beyond). The era effects show that racial disparities in infection risk were not constant, presenting a complex picture of the evolution of COVID infection in San Francisco. The analysis illustrates how Bayesian hierarchical models with flexible time components can recover both persistent and time-varying disparities in infectious disease burden.

1 INTRODUCTION

San Francisco was an early center of the COVID-19 (coronavirus disease 2019) pandemic in California: the first COVID-related death in the United States occurred in neighboring Santa Clara county on February 6, 2020 (NBC Bay Area, 2020), and Bay Area counties were the first in the United States mainland to implement mandatory shelter-in-place orders in March 2020 (Ravani, 2020). However, the impact of COVID-19 infection was not felt uniformly across demographic groups. These disparities may be attributed to a variety of factors, including the distribution of occupations and essential workers, living density and household size, socioeconomic status, health comorbidities, access to health care (including vaccines) and health information, transit use, and other systemic inequities (Mayo Clinic, 2024). Since these disparities can augment existing disparities in health outcomes, their nature and impact continues to be of interest.

San Francisco's open data portal, Data SF (`https://datasf.org`), provides a rich repository of city data sets and records available to the public. The dataset "COVID-19 Cases by Population Characteristics Over Time" (DataSF, 2020) provides 54,000 rows of new and cumulative COVID-19 cases as determined by case interviews, laboratories, and medical providers, timestamped daily from March 2020 through September 2023 and

grouped by race/ethnicity, age, gender, homeless status, and SRO (single-room occupancy[1]) or nursing facility tenancy, among other categories. Although the data set contains multiple demographic dimensions, this report focuses on disparities by race and ethnicity, one of the most persistent and well-documented axes of inequity in COVID-19 outcomes (Faust et al., 2024). Because each demographic attribute is recorded independently (to protect privacy), analyses must be performed within each demographic category separately rather than jointly across intersecting identities. Nevertheless, the size and granularity of this data set make it possible to examine systematic differences in infection patterns across racial and ethnic groups and across time.

The purpose of this project is twofold: (1) to evaluate whether race- and ethnicity-specific infection rates differ meaningfully across the course of the pandemic in San Francisco, and (2) to extend the baseline model through comparisons and refinements. The remainder of this report details the data set and preprocessing pipeline, presents the baseline model and results, and explores potential extensions and comparisons across different phases of the pandemic that further illuminate demographic disparities in COVID-19 infection dynamics in San Francisco.

2 DATA DESCRIPTION AND PREPROCESSING

Table A1 in the Appendix gives the column descriptions for this dataset. To preprocess the data I first filtered to just the data on race/ethnicity, and then removed data on the racial/ethnic groups of "other," "unknown," and "multi-racial"[2]. To reduce the size of the data set, I then aggregated the daily data by week, resulting in around 1100 rows, for which I created group and time indices. Missing data was not an issue except to fill in a few "NA' values for the cumulative case counts with 0 at the earliest rows of the dataset. Figure 1 shows the counts of cumulative and new cases for the filtered race/ethnic groups across the dates in the data set, giving an initial picture. As can be seen, there does appear to be a disparity in case rates across groups in San Francisco. San Francisco was fortunate to have relatively low per capita infection rates for most of the pandemic, but suffered a strong post-vaccine surge in the winter of 2021/2022 (with a noticeable gap in reported cases around Christmas/New Years).

To get a sense of the distribution of daily new case rates by group, I created a kernel density estimation (KDE) plot of the new cases, as shown in Figure 2. Counts data are classically modeled using Poisson or Negative Binomial (NB) distributions. Whereas

1 SROs are a kind of dense affordable housing defined by the San Francisco Housing Code as having six or more "residential guest rooms" that may be attached to shared bathrooms, kitchens, and living spaces (DataSF, 2020).

2 The data set warns about these groups: "The Census definition is likely not exactly aligned with how the City collects this data. For that reason, we do not recommend calculating population rates for these groups." (DataSF, 2020).

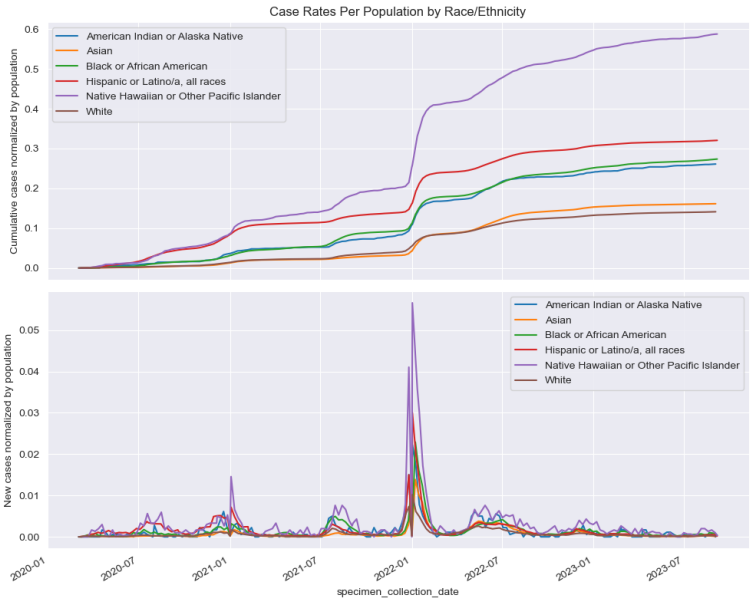


Figure 1—Population-normalized rates of (*top*) cumulative and (*bottom*) new COVID (coronavirus disease 2019) cases in San Francisco, by race/ethnicity.

Poisson requires that the mean and variance are equal, the NB distribution is more flexible in allowing variance to exceed the mean (allows for overdispersion) and is thus a popular choice for modeling disease transmission in epidemiological studies. Thus, an NB distribution is a natural starting point for modeling the data.

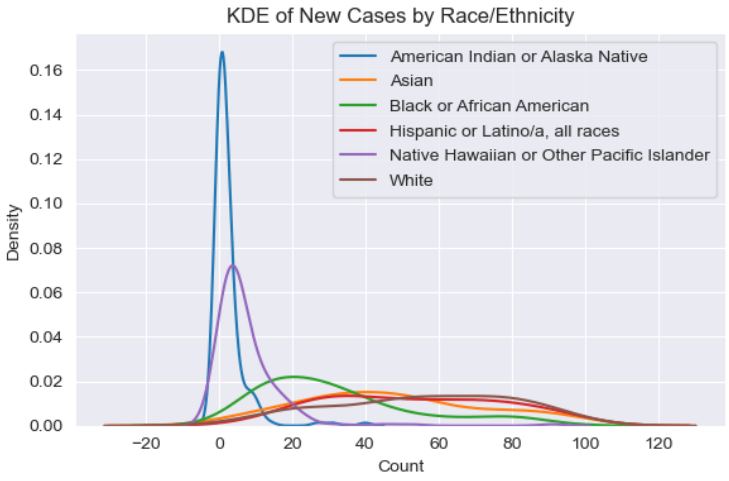


Figure 2—KDE (kernel density estimation) plot of new COVID (coronavirus disease 2019) cases <100 in San Francisco, by race/ethnicity.

However, a simple NB model is insufficient because the data exhibit two identifiable sources of additional structure: time and group (race/ethnicity). Case rates in pandemics can fluctuate dramatically due to changes in behavior, shelter-in-place orders,

mask mandates, vaccines, holidays, new variants, and so-called superspreader events. Any model that fails to account for this temporal dependence risks attributing temporal dynamics incorrectly to demographic differences. Secondly, the raw plots appear to show differences in infection rates across racial/ethnic groups that appear largely persistent across the entire three-year time span, suggesting a group-level effect.

For these reasons, in the next section I present a hierarchical model that includes group-specific effects capturing stable demographic disparities, and a smooth time-varying component capturing shared epidemic trends.

3 BASELINE NEGATIVE BINOMIAL MODEL

3.1 Model Specification

I model the expected case count $\mu_{i,t}$ for each group i in week t as

$$\mu_{it} = \text{population}_i \cdot e^{a_i + f(t)},$$

which includes an exponent to ensure positive counts; a_i represents a group-level effect, and $f(t)$ is a latent smoothed time trend. Equivalently, we write

$$\log \mu_{it} = a_i + f(t) + \log(\text{population}_i).$$

There are several choices for how to model the smoothing parameter f. To start, I chose to model f as a first-order Gaussian Random Walk (GRW1), $f(t) = f(t-1) + \varepsilon_t$, where the so-called innovation ε_t (local drift) is drawn from a normal distribution: $\varepsilon_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_f^2)$. The GRW1 prior functions as a temporal smoother that pulls extreme values towards the local average. This is a common approach in Bayesian epidemiology for time-series count data (see, e.g., Eales et al., 2022, Jersakova, Lomax, and Hetherington, 2022, Steyn et al., 2025). The model of case counts is then

$$\begin{aligned}
y_{it} &\sim \text{NegBin}(\mu_{it}, \phi) \\
\log \mu_{it} &= a_i + f(t) + \log(\text{population}_i) \\
\phi &\sim \text{HalfNormal}(1) \\
a_i &\sim \mathcal{N}(0, \sigma_a) \qquad \left(\text{s.t. } \sum a_i = 0\right) \\
f(t) &\sim \text{GRW1}(\sigma_f) \\
\sigma_a &\sim \text{HalfNormal}(1) \\
\sigma_f &\sim \text{HalfNormal}(1)
\end{aligned}$$

where the a_i are constrained to sum to zero (STZ) to ensure model identifiability. The half-normal priors are intended to be weak but reasonable.

3.2 Model Results

We ran the model in PyMC with 5000 draws each across 4 chains with 1000 burn-in samples and a target accept rate of 0.9. All chains converged with excellent $\hat{r}$ values.

Figure 3 plots the weekly total new observations (all groups) against the posterior predictive mean and 94% highest density interval (HDI) for the posterior draws. This plot shows the effect of the GRW1 in smoothing the weekly spikes in new case counts, so that our model is not able to reproduce the sharpest spikes in the data. Indeed, by comparing the variance in observed new counts data (≈111,174) against the posterior predictive variance (≈24,134), we can see that even with an NB model our sampling is significantly underdispersed relative to the observed case rates.

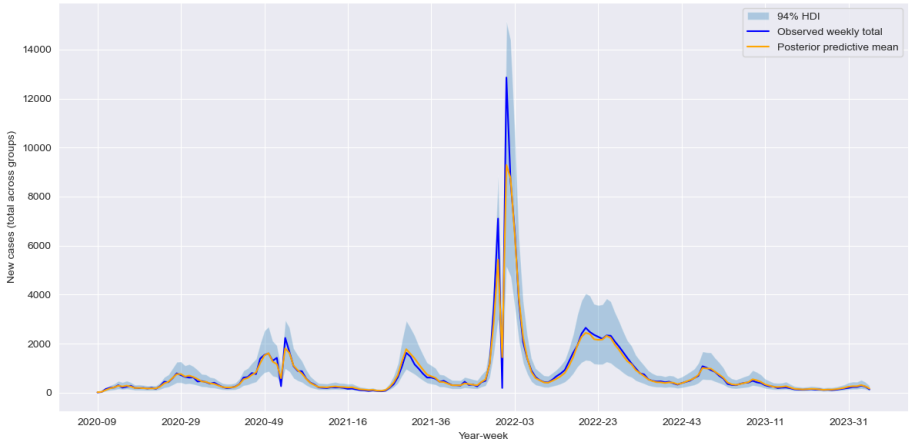


Figure 3—Weekly new case totals (all groups): observed (*blue line*) versus the model's posterior predictive mean (*orange line*) and 95% highest-density interval (HDI).

Figure 4 presents a ridgeplot of the α_i values for each group (which made use of the `contrasts` utility from Redding, 2025), which are also presented in Table 1. As can be seen clearly in the plot and table, Asian and White demographic groups exhibited statistically significant group-level effects that resulted in less case prevalence relative to the overall population, while there was not a statistically significant group-level effect for the American Indian/Alaska Native group. In contrast, Black/African American and Hispanic/Latino groups exhibited statistically significant group-level effects that resulted in heightened COVID case prevalence, and the highest significant group-level effect was exhibited by the Native Hawaiian/Pacific Islander demographic group. Table A2 in the appendix presents the statistically significant pairwise contrasts for the α_i pairs. All pairwise contrasts were statistically significant except for Asian v. White and Black/African American v. American Indian/Alaskan Native. It should be noted that the smallest group in this dataset, the American Indian/Alaskan Native group, comprises less than 1,800 residents, so any data concerning this group should be interpreted with caution, as the small sample size is likely to confer larger stochastic noise.

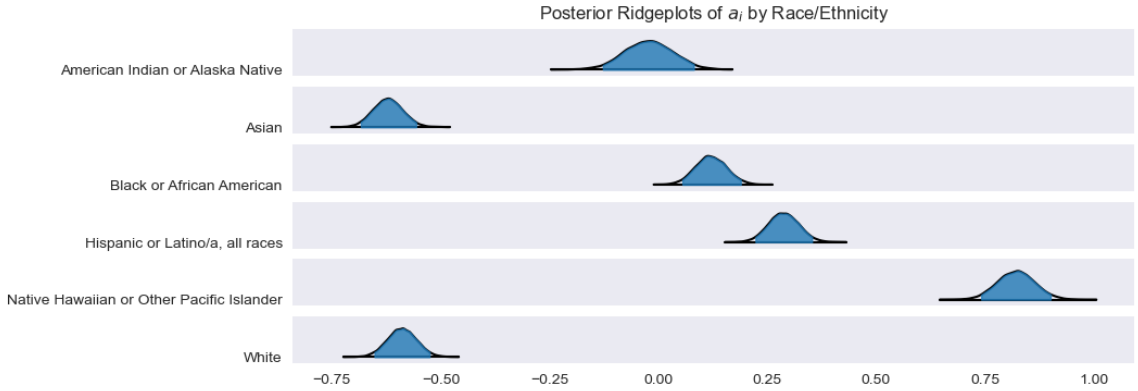


Figure 4—Ridgeplot of the group-level effects a_i under the sum-to-zero-constraint.

Group	Mean	SD	3% HDI	97% HDI
American Indian or Alaska Native	−0.022	0.056	−0.127	0.082
Asian	−0.621	0.034	−0.683	−0.556
Black or African American	0.121	0.036	0.055	0.189
Hispanic or Latino/a (all races)	0.288	0.035	0.223	0.353
Native Hawaiian or Other Pacific Islander	0.822	0.043	0.742	0.901
White	−0.589	0.034	−0.652	−0.525

Table 1—Posterior means and credible intervals for group-level effects a_i.

3.3 Comparison with Poisson and Zero-Inflated Negative Binomial

The comparison of the observed variance in new COVID counts versus the posterior predictive variance of the model suggests that, if anything, the model is underdispersed relative to the data. This gives us evidence that the NB model is likely a better choice than a Poisson would be. Still, for sake of completeness or verification, we also ran a Poisson model for comparison. An additional possibility in the model space is Zero-Inflated Negative Binomial (ZINB), which is an appropriate model when you believe the data is overdispersed (the NB part) and when you think there is an over representation of zeros in the data. This seems possible, as looking at Figure 1, the new cases data do sometime appear to ride the 0 line, and the cumulative cases data appear to be flat at various points, especially at the very beginning of the pandemic. The ZINB model adds to the NB model a logistic regression, which takes a single probability ψ parameter in PyMC, representing the expected proportion of NB draws (rather than zeros). For completeness, we also ran a ZINB model in PyMC, using a mildly informative Beta(10,1) prior on ψ to represent that the zeros are expected to exist at a ratio of 10 to 1 (90.9%). We then compared the three models in PyMC based on their expected log pointwise predictive density (ELPD; Arviz's `compare` function) using the method of leave-one-out

cross-validation (LOO).

The results of these comparisons shown in Figure 5 reveal that Poisson greatly underperforms the NB and ZINB models such that on this scale, the standard error (SE) bands for the NB/ZINB models is not visible. Plotting just the NB and ZINB models alone reveals that their SE bands largely overlap, with our original NB model performing slightly better on average. All models converged, and their $\hat{r}$ values were near 1, but the NB model or ZINB models are clearly the best, with NB slightly better. The posterior value for ψ in the ZINB model was 99.8%, suggesting that the additional structural zeros in the ZINB model were determined to be relatively rare. Indeed, while San Francisco did have remarkably low case rates for much of the pandemic relative to national trends due to early and strict public health measures, there were never weeks of zeros. These results help confirm the original choice of our model, as the ZINB posterior was largely identical to the NB model, and its additional zero inflation didn't improve performance.

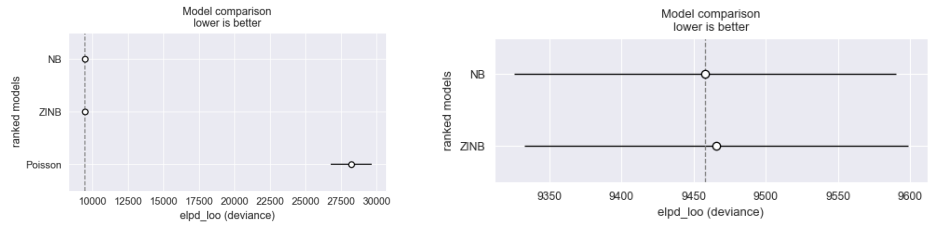


Figure 5—Comparing ELPD-LOO of (*a*) the NB, Poisson, and ZINB models and (*b*) just the NB and ZINB models. Abbreviations: ELPD, expected log pointwise predictive density; LOO, leave-one-out cross-validation; NB, negative binomial; ZINB, zero-inflated NB.

4 COMPARISON OF RACIAL DISPARITIES ACROSS PANDEMIC PHASES

The log-link in our NB model uses a time-invariant group-level effect for race/ethnicity, but were racial disparities really constant across the timeline of the pandemic? Perhaps some groups were affected more strongly by the initial waves of COVID pre-vaccine than the later, post-vaccine waves marked by other COVID variants. If pre- and post-vaccine and the arrival of different variants marked qualitatively different regimes in the data, we might expect to see different posterior distributions. Looking at Figure 1, we see the disparities in cumulative cases look somewhat but not entirely persistent: some groups lines converge or deviate, and some groups seem to be hit especially hard by the post-vaccine Omicron surge in the winter of 2021–2022. Rather than add multiple changepoints to our model to explore this issue, in this section we instead partition the time series into different pandemic eras based on known changes in public health context and dominant variants. These change points will be informed by external knowledge (Proctor, 2021, Ivory et al., 2021, Wikipedia, 2025), rather than inferred solely

from the case series, as in a multi-change point model.

4.1 A Very Brief History of the Pandemic in San Francisco

The time period of our data set begins in March 2020 and ends in September 2023. In the early pandemic, San Francisco took a conservative approach to restrictions on gatherings and businesses, starting with a stay-at-home order on March 16, 2020, that was slowly lifted for different business categories throughout the year (Proctor, 2021). COVID cases went through summer and winter waves, triggering another stay-at-home order in December 2020 as ICU (intensive care unit) capacity at all hospitals reached 0%. The state began vaccinating people in December 2020, starting with the elderly, and rolled out vaccinations to all people aged 16 or older on April 15, 2021. San Francisco mandated proof of vaccination for city workers and patrons of restaurants, gyms, and bars, as did many employers, and was able to achieve one of the highest first-shot vaccinations rates in the country at 86% (Ivory et al., 2021). COVID cases began to decrease statewide and the state lifted most COVID restrictions on June 15, 2021 (Proctor, 2021). Soon after, the Delta COVID variant, a strain twice as transmissible as other strains, began spreading in California, and the case count reached another summer surge. On December 1, 2021, San Francisco recorded the first case in the country of the highly mutated Omicron variant, which was especially transmissible. December 2021 and January 2022 saw by far the largest spike in cases, although the death rate was half of the previous winter's peak. For the remainder of weeks in the dataset, COVID cases exhibited moderate summer and winter surges, and the Omicron variant became the dominant variant throughout California and the world, which it remains to this day (Wikipedia, 2025). The vast majority of pandemic-era orders and restrictions were dropped by late February 2022 (Proctor, 2021), as death rates during the Omicron surge were half that of the previous winter's surge. Although some individuals and businesses continued to keep COVID-era precautions and measures like masking or working from home, increasingly people in the city resumed pre-pandemic activities and behaviors as the population became increasingly vaccinated.

4.2 Model Specification Using Defined Eras

The timeline of the pandemic history suggests at least three qualitatively different transmission regimes: pre-vaccination, from March 2020 to April 2021; public vaccination, from April 2021 to December 2021, a time period that coincided with lifting many COVID restrictions and the emergence of the Delta variant; the Omicron surge and removal of restrictions, from December 2021 through the remainder of the dataset.

To model the changes in disparities across these eras, we partitioned the timeseries according to these time markers and defined an era index $e(t) \in \{1, 2, 3\}$ for each of the

three eras. Our log-link function then becomes

$$\log \mu_{iet} \sim \alpha_i + \beta_e + (\alpha\beta)_{ie} + f(t) + \log(\text{population}_i)$$

where in addition to the α_i, $f(t)$, and $\log(\text{population}_i)$ we had before, β_e is a city-wide era effect and $(\alpha\beta)_{ie}$ is a group $\times$ era interaction term that captures how disparities change by era. We provide weak normal priors on β_e and $(\alpha\beta)_{ie}$:

$$\begin{aligned}
\beta_e &\sim \mathcal{N}(0, \sigma_\beta) \qquad \left(\text{s.t. } \sum \beta_e = 0\right) \\
\sigma_\beta &\sim \text{HalfNormal}(1) \\
(\alpha\beta)_{ie} &\sim \mathcal{N}(0, \sigma_{\alpha\beta}) \qquad \left(\text{s.t. } \sum (\alpha\beta)_{ie} = 0\right) \\
\sigma_{\alpha\beta} &\sim \text{HalfNormal}(0.3)
\end{aligned}$$

where as before, to ensure model identifiability, the β_e and $(\alpha\beta)_{ie}$ are subject to STZ constraints.

4.3 Model Convergence Issues and Revision

We ran the model in PyMC with the same configuration as before: 5000 draws across 4 chains with 1000 burn-in samples and a target accept rate of 0.9. However, this model led to sampling difficulties. Increasing the target accept to 0.97 did not help much, but removing the sum-to-zero constraint for the interaction term and replacing the $\sigma_{\alpha\beta} \sim \text{HalfNormal}(0.3)$ prior with a stronger $\sigma_{\alpha\beta} = 0.2$ value helped somewhat (resulting in as few as 62 divergences and $\hat{r}$ values for f around 1.02). The model was clearly struggling to converge due to the multiple sources of temporal variation. For interpretability and stability, we therefore tried removing the continuous-time random walk, leaving just the discrete era term and the interaction term to model group-by-era interaction. In summary these were our model revisions:

$$\begin{aligned}
\log \mu_{iet} &\sim \alpha_i + \beta_e + (\alpha\beta)_{ie} + \log(\text{population}_i) \\
(\alpha\beta)_{ie} &\sim \mathcal{N}(0, 0.2)
\end{aligned}$$

This yields era-specific log-rates per group and directly addresses whether disparities widened, narrowed, or remained stable over the primary phases of the pandemic. With these changes the model converged and all $\hat{r}$ values were close to 1.00.

4.4 Model Results

The era-level effects β_e from these draws, along with the era-pairwise differences between β_e values, are presented in Table 2. As can be seen in the table, relative to the average across the entire pandemic, the Omicron era exhibited the highest infection risk relative to the citywide average, followed by the pre-vaccine era, with the vaccine rollout and Delta variant presenting the least risk. All pairwise differences between eras were statistically significant. These estimates serve as a useful check that the model captures known historical variation without explicitly encoding it.

Era	Mean	SD	3% HDI	97% HDI
Era 0: Pre-vaccine	−0.227	0.082	−0.382	−0.073
Era 1: Vaccine & Delta	−0.902	0.084	−1.061	−0.744
Era 2: Omicron & beyond	1.129	0.082	0.973	1.281
Era Difference	**Mean**	**SD**	**3% HDI**	**97% HDI**
Era 0 − Era 1	0.6732	0.1442	0.4022	0.9448
Era 1 − Era 2	−2.0290	0.1433	−2.3064	−1.7696
Era 0 − Era 2	−1.3558	0.1403	−1.6253	−1.0970

Table 2—Posterior estimates for era effects β_e and pairwise contrasts between β_e values.

Table 3 presents the posterior differences in group × era interaction parameters for each racial/ethnic group across the three eras, and Figure 6 presents a heat map of the mean values of these differences for clearer visualization. The interaction parameters describe how the infection risk for each racial/ethnic group changed between eras, relative to the average citywide temporal pattern. These effects capture whether a given group experienced disproportionately higher or lower infection pressure in one era compared to another, after adjusting for population size and for the overall differences among eras.

The table and figure show that Hispanic/Latino groups had statistically significant declines in log-risk from Era 0 (pre-vaccine) to Era 1 and Era 2, suggesting that the earliest phase of the pandemic presented a higher infection pressure for this group. This aligns with reporting that Hispanic and Latino communities were overrepresented among essential workers early in the pandemic; indeed, the historically Latino Mission district in San Francisco was an early locus of COVID spread (Wikipedia, 2025).

Another interesting result was that Black/African American residents showed a significant increase from Era 0 to Era 1 and a decrease from Era 1 to Era 2, indicating that this group experienced a comparatively stronger infection pressure in the Delta/vaccine period relative to the early pandemic and Omicron wave relative. This is precisely the *opposite* ordering of era-level infection pressure at the citywide level. Although a full explanation of this finding would go beyond the scope of this report, one speculation might be that Black Americans and Hispanic and Latino groups have reported lower rates of vaccination relative to other groups (Health, 2025); in the absence of a potentially immunity-conferring early-pandemic wave, it is possible that during the less-restrictive Delta/vaccine era, lower vaccination rates among Black residents may have conferred greater risk relative to the overall city population. However, this interpretation is highly tentative, as the present model does not include individual-level vaccination data or

Group	Era Diff	Mean	SD	HDI 3%	HDI 97%
AI/AN	0 − 1	0.270	0.233	-0.171	0.702
	0 − 2	-0.498	0.205	**-0.900**	**-0.123**
	1 − 2	-0.768	0.224	**-1.195**	**-0.358**
A	0 − 1	-0.003	0.187	-0.353	0.345
	0 − 2	-0.309	0.183	-0.640	0.044
	1 − 2	-0.307	0.182	-0.653	0.032
B/AA	0 − 1	-0.431	0.184	**-0.775**	**-0.086**
	0 − 2	0.046	0.186	-0.304	0.397
	1 − 2	0.477	0.185	**0.123**	**0.817**
H/L	0 − 1	0.505	0.183	**0.175**	**0.859**
	0 − 2	0.655	0.181	**0.316**	**0.991**
	1 − 2	0.150	0.186	-0.191	0.502
NH/PI	0 − 1	0.032	0.189	-0.338	0.379
	0 − 2	0.139	0.183	-0.220	0.474
	1 − 2	0.107	0.189	-0.243	0.466
W	0 − 1	-0.342	0.184	**-0.692**	**-0.001**
	0 − 2	-0.091	0.181	-0.438	0.240
	1 − 2	0.251	0.181	-0.081	0.598

Table 3—Posterior differences in group × era interaction effects $(\alpha\beta)_{ie}$ showing the estimated change in log-infection-risk for a given group across pandemic eras. Boldface indicates 94% HDIs that exclude zero. Abbreviations: AI/AN, American Indian or Alaska Native; A, Asian; B/AA, Black or African American; H/L, Hispanic or Latino/a (all races); NH/PI, Native Hawaiian or Pacific Islander; W, white.

direct measures of prior infection-derived immunity. Moreover, white residents also exhibited a statistically significant increase from Era 0 to Era 1, suggesting they also exhibited greater infection pressure in the vaccine/Delta phase, despite relatively higher rates of vaccination. Clearly, a full exploration of the complex reasons why some groups may have experienced increased or decreased infection pressure relative to baseline is going to involve a complex interplay of many different factors and data points that would take us beyond what our dataset can provide.

Asian American and Native Hawaiian/Pacific Islander populations showed no statistically significant changes in their group infection pressure across the different eras, indicating that the temporal pattern of infection risk across eras closely tracked the citywide average. Finally, American Indian/Alaska Native residents showed large shifts with wide intervals, including a significant drop from Era 1 to Era 2. However, because the American Indian/Alaska Native population in San Francisco is quite small (less



Figure 6—Heat map of posterior differences in group × era interaction effects $(\alpha\beta)_{ie}$. Each cell shows the estimated change in log-infection-risk for a given group across pandemic eras. Positive values (*red*) indicate increased infection pressure; negative values (*green*) indicate reductions. Values depicted represent posterior means; non-significant contrasts (94%HDI including zero) are shaded lightly. Abbreviations: AI/AN, American Indian or Alaska Native; A, Asian; B/AA, Black or African American; H/L, Hispanic or Latino/a (all races); NH/PI, Native Hawaiian or Pacific Islander; W, white.

than 1,800 individuals), these effects should be interpreted with caution, as they may be sensitive to small-sample stochasticity.

Together, these results should be interpreted as differences in relative temporal patterns, not absolute differences in case counts or risk. The interaction effects quantify whether a group's infection risk rose or fell more or less than the expected citywide trend when transitioning between major pandemic phases.

4.5 Note About Group Effects in the Era-Effects Model

Because the present era-effects model includes main effects for pandemic era and a full set of era × group interaction effects, the global group-level effects α_i become substantially less interpretable. This is particularly true for the American Indian/Alaska Native population, whose extremely small size (≈1,800 residents) makes its group-level effect much more sensitive to model specification. When interaction terms $(\alpha\beta)_{ie}$ are included, much of the temporal variation that the baseline NB model assigned to α_i is redistributed into era-specific deviations, causing α_i to shift and in this case to falsely appear much more "protective" relative to the citywide average, whereas in the baseline NB model no significant group-level effect was found for this group. For this reason,

we base our substantive interpretation and analysis of the model results primarily on era effects and era–group interaction contrasts rather than on global group effect parameters.

5 DISCUSSION

In this report we have explored the existence of racial and ethnic disparities in COVID case numbers in San Francisco across the timeline of the pandemic. The baseline NB model showed that these disparities are quantifiable: whereas Asian and white residents enjoyed lower infection risk relative to the city average, Black, Native Hawaiian/Pacific Islander, and Hispanic/Latino exhibited a relatively greater infection risk. These patterns are consistent with reporting throughout the pandemic that communities of color experienced disproportionate COVID-19 burdens, especially those with larger proportions of essential workers, higher housing density, and lower access to healthcare.

While our initial model treated these group-level effects as time-invariant, in our second model we found that for some groups, this risk changed across different phases of the pandemic. Certain groups such as Hispanic/Latinos exhibited stronger infection pressure earlier in the pandemic, while for others such as Black/African American and white residents, later phases of the pandemic brought less infection risk relative to previous eras. Other still saw no change across different COVID eras, including, notably, Native Hawaiian/Pacific Islanders, who exhibited the highest infection pressure of all groups, which did not significantly change across the phases of the pandemic. These observations align with broader epidemiological findings that vulnerability is shaped by many complex and intersecting factors that vary across social groups and time.

Limitations of the analysis include the inability to study cross-classified demographic characteristics (e.g., age-by-race), the potential for reporting noise in weekly aggregated case data, and the exclusive use of confirmed case counts rather than testing or hospitalization information. Additionally, the smallest racial groups (e.g., American Indian/Alaska Native) yielded noisier estimates due to very small population denominators.

Future work could incorporate spatial information (e.g., neighborhood-level case counts) either as a separate hierarchical layer or via spatially correlated priors. Mechanistic extensions, such as time-varying reproduction number models or more sophisticated changepoint detection, could also be explored. Combining this modeling approach with vaccination and socioeconomic data would further help explain the divergent trajectories observed across demographic groups.

Despite these limitations, the results demonstrate that hierarchical Bayesian modeling provides a powerful framework for uncovering evolving disparities in infectious disease outcomes. The findings offer insight into how different communities experienced differ-

ent phases of the pandemic and underscore the importance of flexible, data-informed approaches when examining the complex social dynamics of public health crises.

REFERENCES

[1] DataSF (2020). *ARCHIVED: COVID-19 Cases by Population Characteristics Over Time*. URL: https://data.sfgov.org/Health-and-Social-Services/ARCHIVED-COVID-19-Cases-by-Population-Characterist/j7i3-u9ke/about_data.

[2] Eales, O, Ainslie, KEC, Walters, CE, and et al. (2022). "Appropriately smoothing prevalence data to inform estimates of growth rate and reproduction number". In: *Epidemics*, 40:100604. URL: https://doi.org/10.1016/j.epidem.2022.100604.

[3] Faust, JS, Rendon, B, Bongiovanni, T, and et al. (2024). "Racial and ethnic disparities in age-specific all-cause mortality during the COVID-19 pandemic". In: *JAMA Netw. Open*, 7(10):e2438918. URL: https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2824690.

[4] Health, Calif. Dep. Pub. (2025). *COVID-19 vaccination data*. URL: https://www.cdph.ca.gov/Programs/CID/DCDC/Pages/COVID-Vaccine-Data.aspx.

[5] Ivory, D, Smith, M, Lee, JC, and et al. (2021). "See how vaccinations are going in your county and state". In: *The New York Times*, updated Oct. 22, 2022. URL: https://www.nytimes.com/interactive/2023/us/san-francisco-california-covid-cases.html.

[6] Jersakova, R, Lomax, J, and Hetherington, J (2022). "Bayesian imputation of COVID-19 positive test counts for nowcasting under reporting lag". In: *J. R. Stat. Soc. C: Appl. Stat.*, 71(4):834–60. URL: https://doi.org/10.1111/rssc.12557.

[7] Mayo Clinic (2024). "COVID-19 infections by race: What's behind the health disparities?" In: retrieved 23 Nov. 2025. URL: https://www.mayoclinic.org/diseases-conditions/coronavirus/expert-answers/coronavirus-infection-by-race/faq-20488802.

[8] NBC Bay Area (2020). "Autopsy reports date first COVID-19 Bay Area deaths to February". In: *NBC Bay Area*, 22 Apr. URL: https://www.nbcbayarea.com/news/local/south-bay/autopsies-date-first-covid-19-bay-area-deaths-to-february/.

[9] Proctor, R (Mar. 2021). "Remember when? Timeline marks key events in California's year-long pandemic grind". In: *CalMatters*, updated Jul. 31, 2023. URL: https://calmatters.org/health/coronavirus/2021/03/timeline-california-pandemic-year-key-points/.

[10] Ravani, Sarah (2020). "Bay Area coronavirus decision: behind the scenes of nation's first shelter-in-place order". In: *San Francisco Chronicle*, 21 Mar. URL: https://www.sfchronicle.com/bayarea/article/Bay-Area-coronavirus-decision-Behind-the-scenes-15148425.php.

[11] Redding, A. (2025). "9. Simvastatin*". In: *ISYE 6420: Bayesian Statistics*. TA notes, accessed Nov. 19. URL: https://areding.github.io/6420-pymc/unit7/Unit7-demo-factorial.html.

[12] Steyn, N, Chadeau-Hyam, M, Elliott, P, and Donnelly, CD (2025). "A Bayesian model for repeated cross-sectional epidemic prevalence survey data". In: *PLOS Comput. Biol.*, 21(10):e1013515. URL: https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1013515.

[13] Wikipedia (2025). *SARS-CoV-2 Omicron variant*. URL: https://en.wikipedia.org/wiki/SARS-CoV-2_Omicron_variant.

APPENDIX

Table A1. Column names and descriptions for the SF COVID-19 cases dataset (DataSF, 2020)

Column Name	Description
specimen_collection_date	The date the positive test was collected (the date the person was tested).
population_estimate	2016–2020 5-year American Community Survey (ACS) population estimate for a given characteristic type and characteristic group.
new_cases	Cases are counted as confirmed on the date of specimen collection after a positive lab test result.
data_loaded_at	Timestamp when data entered the Socrata system.
data_as_of	Timestamp for last update date in the source system.
cumulative_cases	Cumulative cases as of the specified date.
characteristic_type	Overall topic area for a given population characteristic; these are subgroups or demographic cross-sections, such as age.
characteristic_group_sort_order	Sort order of the characteristic group to aid in visualizing data.
characteristic_group	Each group or category within a characteristic type or topic area (e.g., 0–4 yrs, 5–10 yrs).

Table A2. Posterior differences in group effects α_g for race/ethnicity in the baseline model, showing only statistically significant contrasts (94% HDI excludes 0).

Contrast	Mean	SD	HDI 3%	HDI 97%
AI/AN − A	0.599	0.074	0.458	0.736
AI/AN − H/L	−0.310	0.075	−0.447	−0.166
AI/AN − NH/PI	−0.844	0.080	−0.990	−0.691
AI/AN − W	0.567	0.074	0.425	0.701
A − B/AA	−0.741	0.052	−0.836	−0.641
A − H/L	−0.909	0.051	−1.007	−0.814
A − NH/PI	−1.443	0.059	−1.550	−1.330
B/AA − H/L	−0.167	0.052	−0.265	−0.071
B/AA − NH/PI	−0.701	0.060	−0.815	−0.589
B/AA − W	0.709	0.051	0.615	0.808
H/L − NH/PI	−0.534	0.060	−0.647	−0.422
H/L − W	0.877	0.050	0.782	0.969
NH/PI − W	1.411	0.059	1.300	1.521

Abbreviations: AI/AN, American Indian or Alaska Native; A, Asian; B/AA, Black or African American; H/L, Hispanic or Latino/a (all races); NH/PI, Native Hawaiian or Pacific Islander; W, white.